

CUSTOMER CHURN ANALYSIS AND PREDICTION

Internship Project Report

Submitted By

Prasanta Kumar Nandi

Internship Domain

Data Science

Organization

SaiKet Systems

Project Title

Customer Churn Analysis and Prediction

CERTIFICATE OF COMPLETION

This report presents the work completed as part of the Data Science Internship Program at SaiKet Systems. The project focuses on analyzing customer churn behavior in a telecommunications company and developing machine learning models to predict customers who are likely to discontinue the service.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to SaiKet Systems for providing me with the opportunity to work on this Data Science internship project. This project helped me gain practical exposure to data preprocessing, exploratory data analysis, customer segmentation, machine learning model development, and business recommendation generation.

ABSTRACT

Customer churn is one of the major challenges faced by telecommunications companies. The objective of this project is to analyze customer churn patterns and develop predictive models capable of identifying customers at risk of leaving the company. Various stages including data preparation, exploratory data analysis, customer segmentation, machine learning model building, model evaluation, and business recommendation generation were performed.

The project achieved a prediction accuracy of approximately 82% using Logistic Regression and identified key factors contributing to customer churn.

TABLE OF CONTENTS

1. Introduction
 2. Project Objectives
 3. Dataset Description
 4. Task 1 – Data Preparation
 5. Task 2 – Exploratory Data Analysis
 6. Task 3 – Customer Segmentation
 7. Task 4 – Churn Prediction Model
 8. Task 5 – Model Evaluation and Interpretation
 9. Task 6 – Business Recommendations
 10. Conclusion
 11. References
-

INTRODUCTION

Customer churn refers to the loss of customers who discontinue using a company's products or services. High churn rates can negatively impact revenue, profitability, and long-term business growth.

Telecommunication companies continuously monitor customer behavior to identify factors influencing churn and take preventive actions.

The purpose of this project is to utilize data science techniques to understand churn behavior and predict future churn risks.

PROJECT OBJECTIVES

The main objectives of the project are:

- Analyze customer churn patterns.
 - Explore customer demographics and service usage.
 - Identify high-risk customer segments.
 - Develop machine learning models for churn prediction.
 - Evaluate model performance.
 - Provide actionable business recommendations.
-

DATASET DESCRIPTION

Dataset Name: Telco Customer Churn Dataset

Number of Records: 7043

Number of Features: 21

Target Variable: Churn Important

Features:

- Gender
- Senior Citizen
- Partner
- Dependents
- Tenure
- Contract Type
- Payment Method
- Monthly Charges
- Total Charges

TASK 1 – DATA PREPARATION

Activities Performed:

- Loaded dataset using Pandas.
- Conducted initial data exploration.
- Checked data types and structure.
- Identified missing values.

Converted TotalCharges to numeric format.

-

- Replaced missing values using median imputation.
- Encoded categorical variables using One-Hot Encoding.
- Split dataset into training and testing sets.

Outcome:

A clean and machine-learning-ready dataset was prepared successfully.

TASK 2 – EXPLORATORY DATA ANALYSIS

Analysis Performed:

- Overall churn rate analysis.
- Gender-wise churn distribution. • Partner-wise churn distribution.
- Dependent-wise churn distribution.
- Tenure analysis.
- Contract type analysis.
- Payment method analysis.

Key Findings:

- Month-to-month customers exhibited the highest churn.
 - Electronic check users showed elevated churn rates.
 - New customers were more likely to churn.
 - Long-term customers demonstrated better retention.
-

TASK 3 – CUSTOMER SEGMENTATION

Segmentation Criteria:

-

-
- A. Tenure Segmentation New

Customers (0–12 months)

Medium Customers (13–48 months)

- Loyal Customers (49–72 months)

B. Monthly Charges Segmentation

- Low Value

- Medium Value

- High Value

Key Findings:

- New customers had the highest churn.
- Loyal customers had the lowest churn.
- Medium-value customers showed significant churn activity.
- 578 high-value customers were identified as churned customers.

TASK 4 – CHURN PREDICTION MODEL

Machine Learning Models Used:

1. Logistic Regression

2. Decision Tree

Model Training Process:

- Data splitting
- Feature preparation
-

-
- Model fitting
- Prediction
- Generalization
- Evaluation Metrics:

- Accuracy

- Precision

Recall

F1 Score

TASK 5 – MODEL EVALUATION AND INTERPRETATION

Logistic Regression Results:

Accuracy: 82.11%

Precision: 68.50%

Recall: 60.05%

F1 Score: 64.00%

AUC Score: 86.23%

Decision Tree Results:

Accuracy: 70.97%

Precision: 45.19%

Recall: 45.31%

F1 Score: 45.25%

Model Comparison:

Logistic Regression significantly outperformed the Decision Tree model and was selected as the final predictive model.

Important Features Influencing Churn:

- InternetService_Fiber optic

-
- PaperlessBilling_Yes
- PaymentMethod_Electronic check
- MultipleLines_No phone service
- StreamingMovies_Yes

TASK 6 – BUSINESS RECOMMENDATIONS

Recommendation 1: Encourage customers to switch from month-to-month contracts to annual plans.

Recommendation 2: Promote automatic payment methods instead of electronic checks.

Recommendation 3: Provide onboarding support for new customers.

Recommendation 4: Implement loyalty programs for high-value customers.

Recommendation 5: Improve Fiber Optic service quality and customer support.

Expected Benefits:

- Reduced churn rate
- Improved customer satisfaction
- Increased customer retention
- Enhanced revenue generation

CONCLUSION

The project successfully analyzed customer churn patterns and developed predictive models capable of identifying customers likely to leave the company.

Among the evaluated models, Logistic Regression achieved the highest performance with an accuracy of 82.11% and an AUC score of 86.23%.

The analysis identified several important churn-driving factors including Fiber Optic Internet service, Electronic Check payment methods, and Month-to-Month contracts.

-

- The proposed business recommendations can help the organization reduce churn and improve long-term customer retention.
-

REFERENCES

1. Python Documentation
2. Pandas Documentation
3. Scikit-Learn Documentation
4. Matplotlib Documentation
5. Seaborn Documentation
6. Telco Customer Churn Dataset